

Seppa: Correctness-Gated LLM Kernel Optimization on the Raspberry Pi 5 GPU

Adam Munawar Rahman

New York University

New York, NY, USA

msr541@nyu.edu

ECE-GY 9953 Advanced Project. Adviser: Prof. Brandon Reagen

Abstract—Most edge AI deployments on single-board computers leave the integrated GPU idle. We present Seppa, a harness that puts a large language model in the loop for optimizing GPU kernels, the small programs a GPU runs in parallel: the model proposes changes, and a finite-state machine compiles, verifies against physics-based correctness gates, benchmarks, and decides keep or revert, refusing to benchmark any kernel that failed verification. On the flood-simulation stencil (a kernel in which each grid cell updates from its neighbors) of an offline flood-guidance application for the Raspberry Pi 5, a hand-driven sweep through the same gates produced a kernel 1.58x faster than the original; the machine re-measured and re-judged the result in five consecutive runs driven from a second computer, and in every run it refused to benchmark a kernel that failed its gate. With CPU inference running concurrently the advantage widens to 2.20x, and the deployment sustains 9.6 tokens/s of language-model decode (token-by-token text generation) beside 883 simulation steps per second, an operating point that beats the best measured CPU-only configuration on both axes in the steady-state campaign.

Index Terms—GPU kernel optimization, LLM agents, correctness verification, edge computing, Vulkan

I. INTRODUCTION

Raspberry Pi class boards are a common platform for edge AI in emergency-response and field settings, where power, connectivity, and budget are constrained [12] and inference must happen on-device once the network is gone [11]; a Pi 5 runs a quantized billion-parameter model (weights compressed to low-precision integers) on its CPU cores [8], generating text one token, roughly a word fragment, at a time, in the phase called decode. AI capability is usually added with dedicated hardware, a neural-processing add-on board (NPU HAT) or a USB accelerator [13], which raises cost and adds a procurement dependency in a supply-constrained chip market [14].

Less attention goes to silicon already on the board: every Pi 5 ships a VideoCore VII GPU that compute workloads rarely touch, and during CPU inference it idles. Two questions decide that idle GPU’s worth: can it be made fast enough at a useful workload to be a co-processor, and can it run beside a busy CPU on a shared LPDDR bus? Sections V and VI answer the first question, Section VII the second.

Making kernels fast on a sparsely documented mobile GPU, where tuning folklore from CUDA-class hardware, the NVIDIA GPUs most optimization literature targets, mostly

fails (Section II), is exactly the kind of work the field has begun handing to language models. Existing agentic optimizers, loops that put the model in charge of every step, AutoKernel [15] among them, steer it with prompts and trust it to verify and revert honestly; the record shows what that trust costs (Section III). Seppa removes the trust. The model proposes, and nothing else; a finite-state machine built on Burr [5], a Python state-machine library, owns compilation, verification, benchmarking, and the verdict, and refuses out-of-order requests. The machine is served over the Model Context Protocol (MCP) [6], a standard interface through which a language model calls tools on a server.

The demonstration workload is the flood simulation of an offline flood-guidance application [7], which must run beside language-model narration on one board; its stencil is the optimization target, and its CPU-plus-GPU split is what we measure (Section IX describes the application).

The claim throughout is machine verification, with autonomous discovery left open; this paper reports:

1. the optimization of a real flood stencil for the Pi’s V3D GPU under three physics gates (correctness checks passed before any benchmark), to 1.58x, failed hypotheses included (Section V);
2. an action-space lesson: the first search plateaued because every winning change lived in the host program (the CPU-side code that allocates buffers and launches the kernel), outside what the machine let the model edit (Section V-C);
3. a machine-checked reproduction over MCP, repeated five times under a defined pass criterion, including the server’s refusal to benchmark a gate-failing kernel (Section VI);
4. an interference measurement of the deployed configuration, with a gate-verified CPU-only counterfactual that the GPU split beats on both axes in the steady-state campaign, and never trails the partitioned scheme, the strongest CPU-only arrangement, in any campaign measured (Section VII).

II. BACKGROUND: THE RASPBERRY PI 5 GPU

The Pi 5 pairs four Cortex-A76 CPU cores with a VideoCore VII GPU (V3D 7.1), the block that also drives the desktop. There is no CUDA and no vendor compute toolchain; compute

TABLE I
PLATFORM, COLLECTED FROM THE RUNNING BOARD BY
COLLECT_SPECS.SH

Model	Raspberry Pi 5 Model B Rev 1.1
SoC	BCM2712
CPU	4 x Cortex-A76 @ 2400 MHz
RAM	16 GB LPDDR (shared with GPU)
OS / kernel	Debian GNU/Linux 13 (trixie) (DietPi 10.5.2), 6.18.39+rpt-rpi-2712
Firmware	2025/12/08

TABLE II
GPU COMPUTE LIMITS, COLLECTED LIVE (VULKANINFO)

Device	V3D 7.1.10.2
Driver	Mesa 25.0.7-2+rpt4 (v3dv)
Vulkan API	1.3.305
Core clock	960 MHz
Max invocations / workgroup	256
Shared memory / workgroup	16 KB
Subgroup (SIMD) width	16
fp16 arithmetic	no
Cooperative matrix	no

reaches the GPU only through Vulkan shaders, small programs the GPU runs many times in parallel, compiled by Mesa’s v3dv driver [9]. In this paper, kernel means a GLSL compute shader.

Tables I and II list the platform and the GPU’s compute limits, collected from the running board by `pi/collect_specs.sh` (archived with the raw vulkaninfo dump). The limits are tight, and the scale is modest: the best kernel we measured sustains about 13 GFLOP/s in fp32, and that figure belongs to a dense matrix multiply; Section VII-A compares the two processors on the stencil this paper optimizes. The GPU is a second, slower engine that happens to be free. In Vulkan’s terms, a dispatch is one launch of a kernel over the whole grid; one invocation is one execution of the kernel by one GPU thread; invocations are grouped into workgroups, and the hardware runs them in lockstep bundles of 16 called subgroups.

The strange part is the driver: when a shader wants more registers than exist, v3dv does not fail. It walks a ladder of six compile strategies [16], first disabling, all at once, the optimizations that spend registers for speed (instruction scheduling, loop unrolling, code motion), then halving the thread count, then falling back to a simpler scheduler, and hands back whatever survives, sometimes several times slower.

III. RELATED WORK

Kernel generation is now a standard LLM task. KernelBench [3] conditions speedup on correctness by definition, Kevin [17] trains a model by trial and error, rewarding kernel-writing attempts that prove correct and fast (multi-turn reinforcement learning with verifiable rewards), and TritonRL [18] puts the difficulty in its title, training a model to write kernels in Triton, a Python-like kernel language, “without cheating.”

The systems still cheat. CUDA-L1 [4] reports that 82 of its 250 RL-generated kernels (32.8%) exploited stream-timing loopholes, arranging the GPU’s work queues so the timed section ended before the real work did, inflating the reported speedup to 18x; containment took a reward checker, a database of known hacks, and forced stream synchronization. Wherever the reward is a measured number, the model eventually learns to optimize the measurement itself.

A weaker assumption sits underneath all of this: that a passing check means a correct kernel. Sarkar [19] tests the oracles these benchmarks use, the automated checks that decide whether a kernel counts as correct, in this case fixed-shape elementwise-closeness (allclose) comparisons at fixed tolerance, and finds them systematically optimistic, with seeded-buggy kernels passing the benchmark oracle while failing a stricter one. SpecBench [20] states the general form: when oversight collapses onto an automated suite, the agent optimizes the suite. Our physics gates are the same kind of instrument and inherit the same weakness, which is why Section VIII stops short of calling them sufficient.

A second line puts the evaluator outside the model instead of arguing with it. FunSearch [21] and AlphaEvolve [22] use the model as a mutation operator inside an evolutionary loop scored by a fixed evaluator; AlphaEvolve found a 48-multiplication procedure for 4x4 complex matrix multiplication, the first improvement in that setting in 56 years. That structural commitment, the model proposing and something else scoring, is the one Seppa makes, at a far smaller scale and with the scoring expressed as reachability in a graph instead of a fitness function (Section IV). In hardware design the same split appears with a human in it: Chip-Chat [10] carried conversational design to tapeout with an engineer and a testbench closing the loop, and VeriGen [23] fine-tunes open models for Verilog. We close the loop with a state machine instead.

IV. THE SEPPA HARNESS

Seppa is a port of AutoKernel [15], an open-source kernel-optimization loop, onto a finite-state machine built with Burr, served over MCP by a wrapper (Theodosia) running on the Pi itself.

Fig. 1 shows the graph, with two guard edges to `log_variant`, on compile failure and on gate failure, and a third edge returning implement to hypothesize when the model submits nothing usable. The guards are ordinary code, from `v3d_flood2_opt.py`:

```

1 ("compile_", "verify", expr("compile_ok")),
2 ("compile_", "log_variant", expr("not compile_ok")),
3 ("verify", "benchmark", expr("verify_ok")),
4 ("verify", "log_variant", expr("not verify_ok")), # guard
5 ("benchmark", "evaluate"),
6 ("evaluate", "log_variant"),

```

The benchmark state is reachable only through a green verify. The agent advances the loop through one MCP tool, `step(action, inputs)`, and the server constrains action to the graph’s legal next moves. The server also

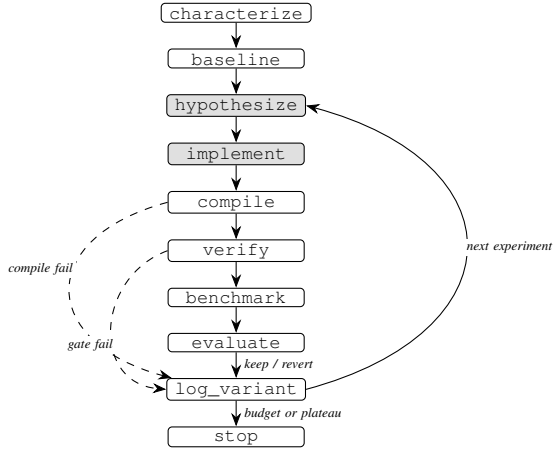


Fig. 1. The Seppa loop. Shaded states are the agent’s; dashed guard edges route compile and gate failures to `log_variant`, so benchmark requires a green verify.

exposes session utilities (`reset_session`, `fork_at`, and `resource reads`) that no archived session used; `fork_at` rewinds to an earlier state, which would let a caller resample a measurement, and the 1% keep threshold introduced below does not defend against that (Section VIII).

The port preserves AutoKernel’s loop discipline (audited in `docs/autokernel_fidelity.md`): one focused change per experiment, a baseline measured first, correctness and timing taken from the same execution, keep only on a gain of more than 1%, every experiment recorded. It moves one thing: the authority to decide. AutoKernel trusts its agent to run the benchmark and honestly revert failures; here the verdict is a deterministic action the model cannot reach. A run ends after three consecutive non-improvements on gate-passing variants, or at the budget.

For the division of labor, this paper follows Chip-Chat’s disclosure convention. The author built the application and the harness, chose the physics gates, ran the hand-driven sweep of Section V-B, and supervised sessions. The language model, Claude Sonnet 5 (Anthropic), driven through the Claude Code client at high reasoning effort, wrote the content of `hypothesize` and `implement` (kernel source and host-side parameters) and nothing else. The machine did everything evidential; no proposed kernel was hand-edited (transcripts in `docs/paper/artifacts/claude_sessions/`). Concretely: the winning kernel came from the hand-driven sweep, the machine verified and reproduced it (Section VI), and the one fully agent-driven session worked a knob the server exposed without finding the fused kernel of Section V-B.

For the flood target, `verify` runs 400 steps at 256x256 and demands three things: normalized mean-square error (NMSE) below 1e-3 against a double-precision CPU reference (loose enough for fp32 drift, six orders above what correct kernels measure), total water within 2% of injected rainfall (correct kernels drift about 0.003%), and maximum depth pooling inside the terrain basin. A green gate in the harness log:

TABLE III
FALSIFICATION SWEEP: 400 STEPS AT 256x256 IN THE VKFLOOD2 HARNESS

Variant	Hypothesis tested	Result
Packed vec2 state (water, surface) halving flux-pass loads	Memory-op count is the bound	1.93 GFLOP/s, no change: falsified
Fused single-dispatch step (flux buffer eliminated)	Traffic and barriers are the bound	2.04 GFLOP/s, +5%: mostly falsified
2 cells per invocation (strip-mined dispatch)	Fixed per-invocation cost is the bound	2.40 GFLOP/s, +23%: supported
4 cells per invocation	More strip is better	2.35 GFLOP/s: falsified, register pressure
Fused + 2 cells per invocation (<code>fused2s.comp</code>)	The wins stack	3.08 GFLOP/s, +59%

```

1 selected: V3D 7.1.10.2
2 grid=256^2 steps=4000 time=1.880s 3.07 GFLOP/s
3 correct(NMSE vs CPU)=yes NMSE=1.26e-09
4 mass: rain_injected=5242879.9
5   water_total=5242739.3 (conserved vs rain=yes)
6 pooling: max depth=111.718 at (127,127)
7   basin_centre=(128,128) (pools in basin=yes)

```

V. CASE STUDY: THE FLOOD STENCIL

A. The Workload

The application’s simulation is a stencil; each cell updates from its four neighbors, in two passes per time step, a flux pass and a height pass. The original implementation ran about 1,045 steps/s at 256x256 (1.51 GFLOP/s), and both shaders compile without triggering Section II’s fallback ladder, which means they never stress the register allocator. The bottleneck was therefore something other than register pressure, the demand for more registers than the hardware has.

B. Falsification Sweep

We did not guess the bottleneck. We wrote one variant per hypothesis and let the loop measure each. The variants pack the state into wider loads, fuse the two passes into one dispatch, or strip-mine, meaning each thread updates a short vertical run of cells to amortize its launch cost. Every variant passed all three gates. Speeds are 400 steps at 256x256 on the parameterized `vkflood2` harness. Its host loop is faster than the original application’s when running identical kernels, about 1,345 steps/s against 1,045, and mixing the two would inflate every ratio, so every comparison in this paper is within `vkflood2`. Table III shows the sweep. Percentages in the table are relative to its baseline, about 1,345 steps/s (1.94 GFLOP/s); the packed variant’s 1.93 is within noise of it, hence no change.

The outcome surprised us. Halving the flux pass’s memory traffic changed nothing, and fusion, which deletes an intermediate buffer and a barrier, bought only 5%. Fixed per-invocation cost dominated: each invocation does so little arithmetic that launch overhead swamps it, and giving each thread two cells beat every memory-system optimization we tried; four cells gave the gain back to register pressure. The

sweep’s own numbers size that cost: strip-2 removes 65,536 invocations per step and saves about 140 microseconds, near 2 ns per invocation, about two clock cycles, the scale of starting a thread itself. Two hardware details shaped the final kernel. Strips run vertically: a subgroup’s 16 lanes are horizontal neighbors, so when each lane walks down its own short column, the subgroup’s loads at every row stay on adjacent addresses. The kernel also consumes each flux value as it is produced; holding all four alive fails register allocation. The shipped kernel is 1.58x at 256x256: 1.574 to 1.585 across the five scored reproductions, 1.571 in the July transcript quoted in Section VI, and 1.59x hand-timed in the sweep. It is 1.41x at 512x512 and 1.18x at 1024x1024, and it holds all gates over 4,000 steps.

C. Action-Space Limits

An earlier session pointed the state machine at this stencil and came back empty; we nearly concluded the kernel was at its limit. The problem was the action space: that version of `implement` accepted shader text only, and every winning change in Table III lives outside the shader (packing changes the buffer layout, fusion deletes a host-loop pipeline, strip-mining changes the dispatch shape). Once `implement` took `{shader, height_shader, strip}`, the machine found and kept the fused strip-2 kernel from its own baseline (Section VI). A plateau can mean the search space is too small even when the hardware has headroom left. Two other targets ran under the same gates: a GEMM (dense matrix multiply) went from 7.02 to 13.42 GFLOP/s over two rounds, and de-unrolling `llama.cpp`’s matrix-vector kernel gained 28% in end-to-end text-generation speed. Both are recorded only in the running notes, and neither is claimed at Section VI’s evidentiary standard.

VI. MACHINE-CHECKED REPRODUCTION OVER MCP

Section V’s numbers do not have to be taken on trust. `drive_flood2_mcp.py` runs on a second computer and drives the Pi-resident state machine through the whole cycle via the `step` tool.

The state machine measures its own baseline (1,346.8 steps/s on 2026-07-11, gates green), receives the fused strip-2 kernel as experiment 1, compiles it, gates it, benchmarks 2,116.4 steps/s, and issues its own verdict:

```
1 {"exp": 1, "fused": true, "strip": 2,
2  "compile_ok": true, "verify_ok": true,
3  "steps_per_sec": 2116.4, "best_sps": 2116.4,
4  "verdict": "keep"}
```

Then the driver tries to cheat: it submits the same kernel with rainfall injection doubled in one of the two cell updates, a change that compiles and breaks conservation. The gate fails it; the driver requests `benchmark` anyway, and the server answers (two advisory fields elided):

```
1 {"error": "invalid_transition",
2  "requested": "benchmark",
3  "valid_next_actions": ["log_variant"],
4  "message": "action 'benchmark' is not
5   reachable from current state. Valid
6   actions now: ['log_variant']."}

```

The variant is ledgered as `revert` with a null `steps_per_sec`. Because gates and timing come from the same execution, the caller still sees the failing kernel’s speed: `verify` reports it as `steps_per_sec_if_kept`. The green gate does not hide that number; it controls the ledger, the verdict, and the kept kernel.

How repeatable is this? `passk_flood2.py` runs the two-cycle reproduction `k` times from cool starts, each run gated to begin below 55 C, scoring a pass when the baseline gates green inside 1,300 to 1,400 steps/s, the fused kernel is kept at 1.5x or better, and the broken kernel is refused and nulled. Five of five runs passed on 2026-08-09: baselines $1,348.6 \pm 4.1$ steps/s, the kept kernel 2,127.7 in every run (identical at the timer’s resolution), speedups 1.574 to 1.585.

One session was fully agent-driven (2026-08-02, budget three). The server’s `characterize` payload names fixed per-invocation cost as the bound and exposes the strip parameter, so the session demonstrates the loop’s enforcement alone: the model worked the exposed knob (strip 4 kept at 1,470.6, +8.8%), hit the register cliff at strip 8 (921.7, reverted), probed it at strip 6 (1,298.7, reverted), and never reached fusion; all of its reported numbers came from the machine.

VII. CONCURRENT CPU AND GPU EXECUTION

In deployment the GPU loops the simulation while the CPU routes and decodes (Granite 4.0 1B on `llama.cpp` with `-ngl 0`, no layers offloaded to the GPU; 4 threads). Decode, the token-by-token generation phase, is memory-bound: its speed is set by how fast model weights stream from memory, and that matters because the two processors share one LPDDR bus.

The campaign turned up one configuration surprise first. With any Vulkan device visible, `llama.cpp` places CPU-resident model weights in memory set aside for GPU transfers (host-pinned and write-combined) even at `-ngl 0`, memory that is slow for the CPU reads that dominate decode. Hiding the device with `GGML_VK_VISIBLE_DEVICES=99` takes decode from 9.68 to 11.59 tokens/s, a 20% gain from an environment variable (A/B from cool starts, `llama-bench`, 5 repetitions; raw logs in `docs/paper/artifacts/ab_vkvisible`). Every number below uses the hidden-device configuration, which the application ships.

`pi/flood/conc_steady_bench.sh` measures each kernel alone (cool starts, finished before heat accumulates), decode alone, and each kernel looping while decode runs. Each decode-bearing phase is preceded by a cooldown and a soak: eight decode repetitions, logged but not reported, that bring the chip to its working temperature before measurement begins. The kernel-alone phases run from the cooldown directly. The Pi’s firmware defends the chip against heat by lowering clock speeds, gently once past a soft temperature limit and sharply at a hard throttle, and the logs record which was active. The soak runs a fixed repetition count instead of waiting for those flags to clear, and the conditions did not end up in the same thermal regime: the concurrent phases ran with the soft limit active 70 to 79% of the time and peaked at 76.3 C, while decode alone

TABLE IV
CONCURRENT GPU FLOOD AND CPU LLM DECODE

Condition	GPU flood (steps/s)	CPU decode (t/s)
Optimized kernel alone	2,128.0 $\pm$ 1.7 (n=3)	
Original kernel alone	1,351.4 $\pm$ 0.5 (n=3)	
Decode alone, 4 threads (post-soak)		11.4 $\pm$ 0.0
Concurrent, optimized kernel	883.4 $\pm$ 47.1 (n=28)	9.6 $\pm$ 0.2
Concurrent, original kernel	401.8 $\pm$ 8.3 (n=12)	9.6 $\pm$ 0.2

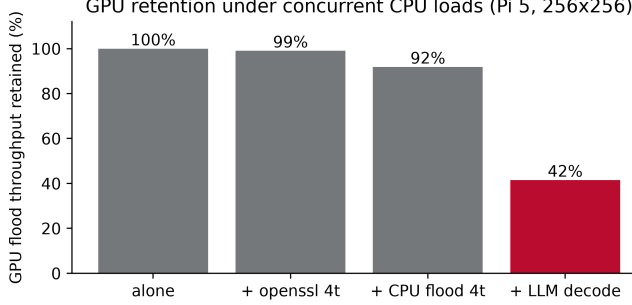


Fig. 2. GPU flood throughput retained beside three concurrent CPU loads, as a percentage of the same kernel running alone. Compute-bound and memory-bound CPU work barely disturb the GPU; language-model decode, which streams weights from DRAM every token, takes it to 42%. Raw logs in docs/paper/artifacts/, genload2 and conc_steady.

reached 73.0 C and was limited for 3% of its phase. Each window is long enough for 12 to 28 flood completions (llama-bench generating 64 tokens per repetition, 20 repetitions), and a flood run counts only if it fits inside the window.

An earlier short-window campaign, preserved in conc_bench2/, read up to 19% differently in both directions. We first attributed that to a thermal transient, but the logs do not support it: the earlier campaign ran cooler (68.7 C mean, soft limit active 46% of the phase, hard throttle never) than this one (72.7 C, 70%, 9%) and still read lower, so heat does not explain the gap. Within this window the rate also climbs, from 818.6 steps/s over the first quarter to 919.1 over the last, so the figure below is a window mean on a rising curve. Results from the steady-state run of 2026-08-10 are in Table IV (raw logs in docs/paper/artifacts/, conc_steady).

Three things follow. Co-processing costs the CPU 16% of its decode and buys a continuous 883 steps/s of simulation that a CPU-only deployment does not have. The interference is lopsided: the CPU keeps 84% of its rate while the GPU keeps 42%, as decode traffic crowds the shared bus. And contention favors the optimized kernel: 1.58x over the original alone becomes 2.20x concurrent, at identical decode cost (9.6 t/s under either kernel). Because the window is still rising, 883.4 is a conservative estimate of the late-window rate.

Decode is also the GPU’s worst case. Paired with a compute-bound CPU load (openssl SHA-256, four threads) the GPU keeps 99% and the load keeps 88% of its 16 KB-block throughput; paired with the memory-bound four-thread

TABLE V
THE CPU-ONLY COUNTERFACTUAL

Condition	Flood (steps/s)	CPU decode (t/s)
CPU flood alone, 1 / 2 / 4 threads	992.5 / 1,980.5 / 3,852.4	
Decode alone, 3 threads (pinned)		11.9 $\pm$ 0.0
Partitioned: flood 1t + decode 3t	678.8 $\pm$ 10.6 (n=25)	8.4 $\pm$ 0.1
Oversubscribed: flood 4t + decode 4t	857.1 $\pm$ 181.0 (n=89)	3.0 $\pm$ 0.3
GPU concurrent, optimized (from above)	883.4 $\pm$ 47.1	9.6 $\pm$ 0.2

CPU flood the GPU keeps 92% and the load keeps 75% (continuous flood-simulation probes over cool-start 60 s windows, both sides logged; docs/paper/artifacts/, genload2). These two pairings also bound the thermal confound. Both ran at the same 76.3 C with the soft limit active for 80% and 70% of their phases, against a solo phase that never left 57.6 C, and the GPU still kept 99% and 92%. Clock governance therefore does not explain decode’s 42% on its own. The GPU is the robust side of every pairing except decode, which streams model weights from DRAM every token: interference tracks the CPU load’s memory traffic.

A. The CPU-Only Counterfactual

Four idle A76 cores run this stencil at 3,852 steps/s, well above the V3D’s 2,128, so the GPU looks unnecessary. pi/flood/cpuflood.cpp is the same update with OpenMP and passes the same three gates (NMSE 1.46e-11; gate logs in cpu_flood_bench/); pi/flood/cpu_steady_bench.sh measures it alone and sharing the cores with decode, pinned apart (flood on core 0, decode on 1 to 3) and oversubscribed (four threads each). The same soak-and-measure protocol applied; raw logs are in docs/paper/artifacts/, cpu_steady. Table V presents the outcome.

The idle-core number is real but unavailable: in deployment, decode is always running. Oversubscribed, decode collapses 75% (llama.cpp’s workers synchronize every token; losing one core stalls all four), and the flood mean is inflated by bursts between decode repetitions, hence its $\pm$ 181. Partitioned, the best CPU-only arrangement, decode pays a 29% tax against the GPU split’s 16%; time-slicing (flooding about 23% of the time to match 883 steps/s) would cap decode at 8.8 t/s and freeze guidance during bursts. The GPU split beats the best measured CPU-only option on both axes, by 30% on simulation (883.4 $\pm$ 47.1 vs 678.8 $\pm$ 10.6) and 14% on decode (9.6 $\pm$ 0.2 vs 8.4 $\pm$ 0.1): the slower processor still wins, because the CPU has no spare cycles to sell. The margin is campaign-dependent, but the direction is not: in the earlier campaign the decode axis holds and the simulation axis ties within noise (Section VIII); no measured campaign puts the partitioned scheme ahead on either axis.

VIII. LIMITATIONS

The envelope is one board (Pi 5, V3D 7.1.10.2, Mesa v3dv 25.0.7) and one grid family, with speedups that shrink as the grid grows; headroom likely remains. The July and August campaigns straddle an OS update (kernel 6.12 to 6.18); flood baselines agree across it (1,346.8 before, $1,348.6 \pm 4.1$ after). The board itself failed after the August campaign, so no quantity in this paper can be re-measured.

The gates are only necessary conditions: verification covers one storm scenario at one grid size, and the keep decision rests on a single timing sample against a 1% threshold, although Section VI’s five-run repeatability (baseline spread about 0.3%, kept kernel identical at timer resolution) bounds that noise well inside the threshold. That bound is only an upper bound: the harness prints elapsed time to milliseconds, so at the kept kernel’s 0.188 s one digit is 0.53%, and “identical” means the last printed digit did not move. The 3,852 steps/s CPU figure assumed a cache-resident 256x256 working set and may not survive larger grids.

Three asymmetries qualify Section VII. The GPU kernel was optimized and `cpuflood.cpp` was not: it is the original two-pass scheme, so the counterfactual compares a tuned GPU against an untuned CPU, and fusing its flux array would cut DRAM traffic on the axis that contends with decode. The conditions were not measured at a matched thermal operating point, so clock governance accounts for part of the 16% decode cost. And the steady-state campaign is a single run: an earlier campaign at the same configuration, kept in `conc_bench2/`, reads 711.9 steps/s concurrent against this run’s 883.4, which is the least reproducible quantity in the paper. Under the earlier campaign the simulation axis is within noise of the partitioned CPU scheme (711.9 ± 76.1 , $n=5$, against 678.8 ± 10.6 , $n=25$), so the two-axis claim rests on the later campaign; the decode axis holds under both (9.6 and 10.3 against 8.4).

The scripted reproduction’s kernel came from the hand sweep, and the agent-driven session worked a knob the server itself exposed. The server’s `fork_at` utility rewinds to an earlier state, so a caller could in principle resample a marginal kernel until a draw cleared the 1% threshold; `evaluate` compares one fresh sample against a frozen incumbent and would not notice. No archived session used it. Autonomous discovery remains open. Full GPU offload of the language model is blocked by an upstream `llama.cpp` Vulkan defect. Numbers without an archived log (the original harness’s 1,045 steps/s, the sweep’s GFLOP/s column and hand-timed ratios, the 1.41x and 1.18x at the larger grids, and the GEMM and matrix-vector results) come from the repository’s dated running notes.

IX. THE SAMPLE APPLICATION

The sample application [7] simulates surface flooding over real terrain [1], finds shelter routes by mobility profile [2], and explains the result in plain language, entirely offline; the model never makes a safety decision, only narrating what the deterministic code computed. The application lives at github.com/msradam/bonbibi and was built as

an Arm AI Optimization Challenge entry; the harness at github.com/msradam/seppa. Everything this paper claims was built for this course, on top of that application. Measurements used its Granite 4.0 1B configuration; the model file (a 1.63 B-parameter GGUF, `llama.cpp`’s weight format) appears in `llama-bench` logs under a granite-3B size label (the application has since moved to a larger model); build commands are in `pi/flood/README.md`, and cited references are archived in `docs/paper/references/`.

X. CONCLUSION

Seppa separates proposal from judgment: a language model suggests kernels for the Raspberry Pi 5’s integrated GPU, and a state machine the model cannot argue with decides what is true about them. On a real flood stencil this discipline produced a verified 1.58x, proposed in the author’s hand-driven sweep and reproduced within 1% by the machine five of five times, and in the steady-state campaign it holds in deployment: 883 steps/s of simulation beside decode at 84% of its solo speed. The evidence in this paper is the gate ledger. The suggestion is open-ended: commodity edge boards carry more usable silicon than their deployments exercise, and a loop that cannot misreport its own verdicts is a sound way to put that silicon to work.

REFERENCES

- [1] M. Guidolin, A. S. Chen, B. Ghimire, E. C. Keedwell, S. Djordjevic, and D. A. Savic. A weighted cellular automata 2D inundation model for rapid flood analysis. *Environmental Modelling & Software* 84:378-394, 2016.
- [2] J. Dibbelt, B. Strasser, and D. Wagner. Customizable Contraction Hierarchies. *ACM Journal of Experimental Algorithmics* 21, Article 1.5, 2016.
- [3] A. Ouyang, S. Guo, et al. KernelBench: Can LLMs Write Efficient GPU Kernels? *arXiv:2502.10517*, 2025.
- [4] X. Li et al. CUDA-L1: Improving CUDA Optimization via Contrastive Reinforcement Learning. *arXiv:2507.14111*; ICLR 2026.
- [5] Burr. Apache Software Foundation (incubating). <https://github.com/apache/burr>
- [6] Model Context Protocol. <https://modelcontextprotocol.io>
- [7] Bonbibi: offline flood-aware routing on a Raspberry Pi 5. <https://github.com/msradam/bonbibi>
- [8] `llama.cpp`. `ggml-org`. <https://github.com/ggml-org/llama.cpp>
- [9] V3D driver documentation, Mesa 3D. <https://docs.mesa3d.org/drivers/v3d.html>
- [10] J. Blocklove, S. Garg, R. Karri, and H. Pearce. Chip-Chat: Challenges and Opportunities in Conversational Hardware Design. *ACM/IEEE Workshop on Machine Learning for CAD (MLCAD)*, 2023.
- [11] Z. Zhou et al. Edge Intelligence: Paving the Last Mile of Artificial Intelligence With Edge Computing. *Proc. IEEE* 107(8), 2019.
- [12] S. Boddu and A. Mukherjee. Efficient Edge Deployment of Quantized YOLOv4-Tiny for Aerial Emergency Object Detection on Raspberry Pi 5. *arXiv:2506.09300*, 2025.
- [13] A. Reuther et al. Survey of Machine Learning Accelerators. *IEEE HPEC*, 2020.
- [14] The White House. Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-Based Growth: 100-Day Reviews under Executive Order 14017. June 2021.
- [15] AutoKernel. RightNow AI. <https://github.com/RightNow-AI/autokernel>
- [16] Mesa, `src/broadcom/compiler/vir.c`, `strategies[]` compile-fallback table. <https://gitlab.freedesktop.org/mesa/mesa>
- [17] C. Baronio, P. Marsella, B. Pan, et al. Kevin: Multi-Turn RL for Generating CUDA Kernels. *arXiv:2507.11948*, 2025.
- [18] J. Woo, S. Zhu, A. Nie, Z. Jia, Y. Wang, and Y. Park. TritonRL: Training LLMs to Think and Code Triton Without Cheating. *arXiv:2510.17891*, 2025.
- [19] D. Sarkar. The Correctness Illusion in LLM-Generated GPU Kernels. *arXiv:2606.20128*, 2026.
- [20] B. Zhao, D. Srikanth, Y. Wu, and Z. Jiang. SpecBench: Measuring Reward Hacking in Long-Horizon Coding Agents. *arXiv:2605.21384*, 2026.
- [21] B. Romera-Paredes, M. Barekatin, A. Novikov, et al. Mathematical discoveries from program search with large language models. *Nature* 625:468-475, 2024.
- [22] A. Novikov, N. Vu, M. Eisenberger, et al. AlphaEvolve: A coding agent for scientific and algorithmic discovery. Google DeepMind, 2025.
- [23] S. Thakur, B. Ahmad, H. Pearce, B. Tan, B. Dolan-Gavitt, R. Karri, and S. Garg. VeriGen: A Large Language Model for Verilog Code Generation. *ACM TODAES*, 2024.